PtoP Supplementary MATERIAL

1 User Study Interface

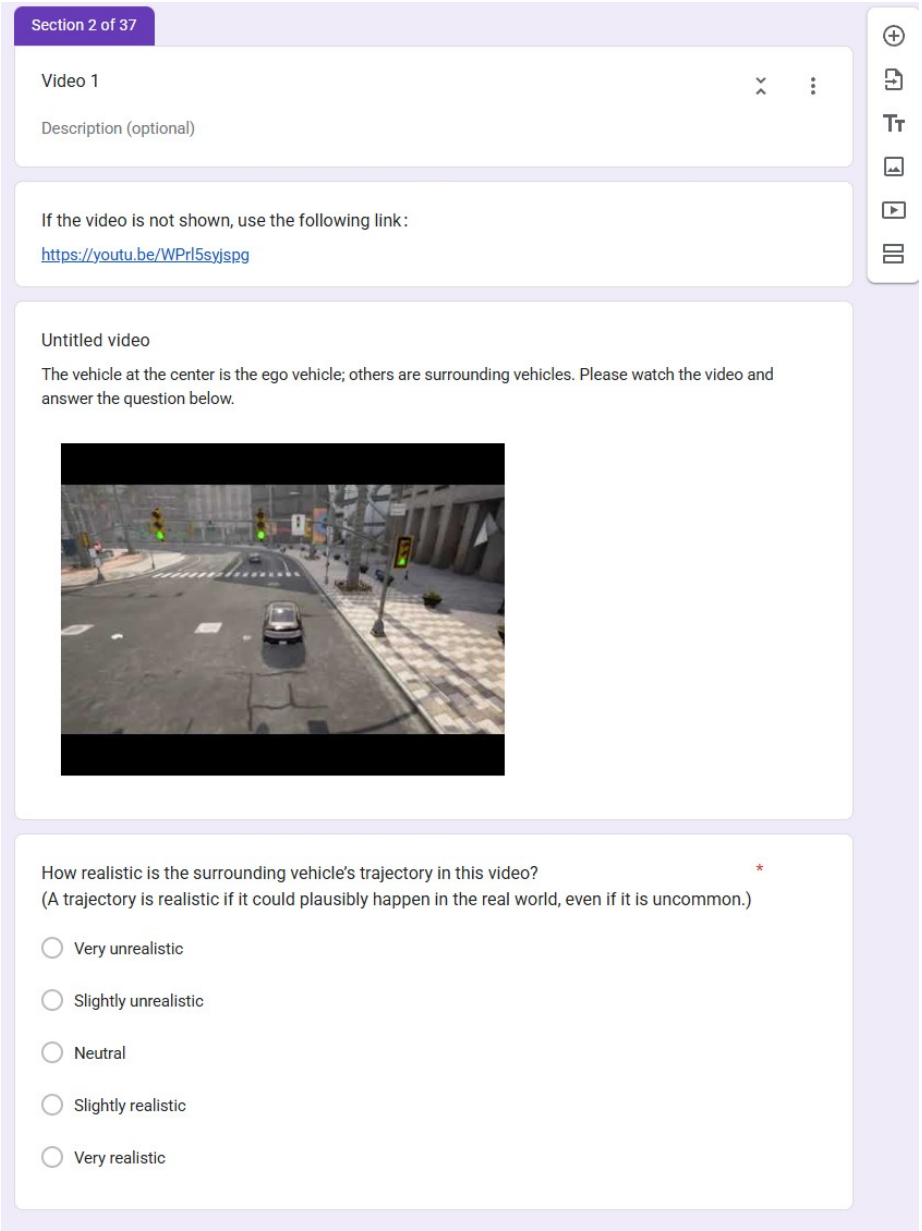

Figure 1: User Study Interface

2 Weighted Fleiss' Kappa Calculation

In our rating scheme, we acknowledge that larger rating discrepancies should correspond to greater disagreement. To account for this, we introduce the Weighted Fleiss' Kappa calculation. We first transfer our Likert scale rating category from text value to numerical value:

- *"Very unrealistic"* is assigned the value 1.
- *"Slightly unrealistic"* is assigned the value 2.
- *"Neutral"* is assigned the value 3.
- *"Slightly realistic"* is assigned the value 4.
- *"Very realistic"* is assigned the value 5.

Definition of the Weight Matrix

Since ratings are ordinal, we apply a quadratic weight function to penalize large discrepancies more than small ones. The weight matrix w_{ij} is defined as:

$$w_{ij} = 1 - \frac{(i-j)^2}{(k-1)^2} \tag{1}$$

where:

- i, j are rating categories $(1, 2, \ldots, k)$.
- k is the total number of rating categories.
- The further apart the categories, the smaller the weight.

In our case, the $k = 5$, an example of the weight matrix is:

$$W = \begin{bmatrix} 1.00 & 0.75 & 0.44 & 0.19 & 0.00 \\ 0.75 & 1.00 & 0.75 & 0.44 & 0.19 \\ 0.44 & 0.75 & 1.00 & 0.75 & 0.44 \\ 0.19 & 0.44 & 0.75 & 1.00 & 0.75 \\ 0.00 & 0.19 & 0.44 & 0.75 & 1.00 \end{bmatrix} \tag{2}$$

Observed Agreement

For each subject (e.g., a video being rated), let f_{ij} be the number of raters assigning a rating of j. The total number of ratings for a given subject i is:

$$p_i = \sum_{j=1}^{k} f_{ij} \tag{3}$$

The observed agreement for subject i is calculated as:

$$P_i = \frac{\sum_{j=1}^{k} \sum_{l=1}^{k} w_{jl} \cdot f_{ij} \cdot f_{il}}{p_i(p_i - 1)} \tag{4}$$

where:

- f_{ij} is the count of raters giving category j to subject i.
- f_{il} represents the number of raters who assigned rating l to subject i.
- w_{jl} is the weight between rating category j and l.
- $p_i(p_i - 1)$ is the total number of rating pairs.
- If $p_i = 1$, then P_i is set to 0 (as no agreement is possible).

The overall observed agreement across all subjects is:

$$\bar{P} = \frac{1}{N} \sum_{i=1}^{N} P_i \tag{5}$$

where N is the total number of subjects.

Expected Agreement

The proportion of ratings assigned to each category across all subjects is:

$$p_j = \frac{\sum_{i=1}^{N} f_{ij}}{\sum_{i=1}^{N} p_i} \tag{6}$$

The expected agreement, assuming ratings were randomly assigned, is:

$$P_e = \sum_{j=1}^{k} \sum_{l=1}^{k} w_{jl} \cdot p_j \cdot p_l \tag{7}$$

Computation of Weighted Fleiss' Kappa

The final formula for the weighted Fleiss' Kappa is:

$$\kappa_w = \frac{\bar{P} - P_e}{1 - P_e} \tag{8}$$

where:

- $\bar{P}$ is the observed agreement.
- P_e is the expected agreement.
- If $\bar{P} = P_e$, then $\kappa_w = 0$, indicating no agreement beyond chance.

Interpretation of Weighted Fleiss' Kappa

The interpretation of κ_w is as follows:

- $0.00 - 0.20$: Slight agreement
- $0.21 - 0.40$: Fair agreement
- $0.41 - 0.60$: Moderate agreement
- $0.61 - 0.80$: Substantial agreement
- $0.81 - 1.00$: Almost perfect agreement

For example, if we compute:

$$\kappa_w \approx 0.714 \tag{9}$$

this suggests **substantial agreement** among raters.

3 Online Control From a Hazard Model

State and control. We represent the dynamic scene by stacking the states of all moving objects $z = \left[s^{(1)}, \ldots, s^{(N)}\right] \in \mathbb{R}^d$. For **vehicles/bicycles** we use a simplified bicycle model:

$$s^{\text{veh}} = (x,\, y,\, \psi,\, v), \qquad u^{\text{veh}} = (a,\, \delta),$$

where (x, y) is position, ψ heading, v longitudinal speed, a longitudinal acceleration (from throttle) and δ steering angle. For **pedestrians**:

$$s^{\text{ped}} = (x,\, y), \qquad u^{\text{ped}} = (u_x,\, u_y),$$

where (u_x, u_y) are longitudinal/lateral displacement (or velocity) commands over one control period.

Single-step discrete dynamics. Let Δt be the control period.
Vehicle/Bicycle (Euler discretization, wheelbase L):

$$x^+ = x + \Delta t\, v \cos\psi, \tag{10}$$
$$y^+ = y + \Delta t\, v \sin\psi, \tag{11}$$
$$\psi^+ = \psi + \Delta t\, \frac{v}{L} \tan\delta, \tag{12}$$
$$v^+ = v + \Delta t\, a, \tag{13}$$

i.e., $s^+ = f_{\text{veh}}(s, u)$.
Pedestrian:

$$x^+ = x + \Delta t\, u_x, \tag{14}$$
$$y^+ = y + \Delta t\, u_y, \tag{15}$$

i.e., $s^+ = f_{\text{ped}}(s, u)$.

Stacking all objects yields the scene update

$$z^+(U) = F\big(z,\, U\big), \qquad U = \big[u^{(1)}, \ldots, u^{(N)}\big]. \tag{16}$$

Objective from a hazard model. Let $h_\theta(\cdot)$ be a differentiable hazard model scoring the next-step scene:

$$J(U) \;=\; h_\theta\big(z^+(U)\big). \tag{17}$$

add a small smoothness regularizer:

$$J_\lambda(U) \;=\; h_\theta\big(z^+(U)\big) \;-\; \lambda \sum_{i=1}^{N} \big\| u^{(i)} - u^{(i)}_{\text{prev}} \big\|^2. \tag{18}$$

One-step ascent and projection. For each object i, perform a parallel one-step ascent with projection onto the feasible control set $\mathcal{U}_i$ (bounds on a, δ or on u_x, u_y):

$$u^{(i)} \;\leftarrow\; \Pi_{\mathcal{U}_i}\Big[u^{(i)} \;+\; \eta\, \nabla_{u^{(i)}} J(U)\Big]. \tag{19}$$

By the chain rule,

$$\nabla_{u^{(i)}} J(U) \;=\; \underbrace{\Big(\nabla_{s^{(i)}} h_\theta\big(z^+\big)\Big)}_{\text{hazard sensitivity}} \underbrace{\Big(\nabla_{u^{(i)}} f_i\big(s^{(i)}, u^{(i)}\big)\Big)}_{\text{dynamics Jacobian}}. \tag{20}$$

Convenient Jacobians (per-step). From the dynamics above, useful partials are:

$$\frac{\partial s^+}{\partial a} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \Delta t \end{bmatrix}, \qquad \frac{\partial s^+}{\partial \delta} = \begin{bmatrix} 0 \\ 0 \\ \Delta t\, \dfrac{v}{L}\delta \\ 0 \end{bmatrix} \quad \text{(vehicle/bicycle)}, \tag{21}$$

$$\frac{\partial s^+}{\partial u_x} = \begin{bmatrix} \Delta t \\ 0 \end{bmatrix}, \qquad \frac{\partial s^+}{\partial u_y} = \begin{bmatrix} 0 \\ \Delta t \end{bmatrix} \quad \text{(pedestrian)}. \tag{22}$$

4 Failure Taxonomy

Figure 2 presents the failure taxonomy. From the results, we find that most safety violations are caused by collisions. Due to the characteristics of the maps, only Town 1 includes red-light violations.

Regarding collisions, Town 1 and Town 7 mainly consist of single-lane roads, so rear-end collisions are more common. In contrast, maps with more multi-lane roads (Town 04 and Town 10) show a higher proportion of side collisions.

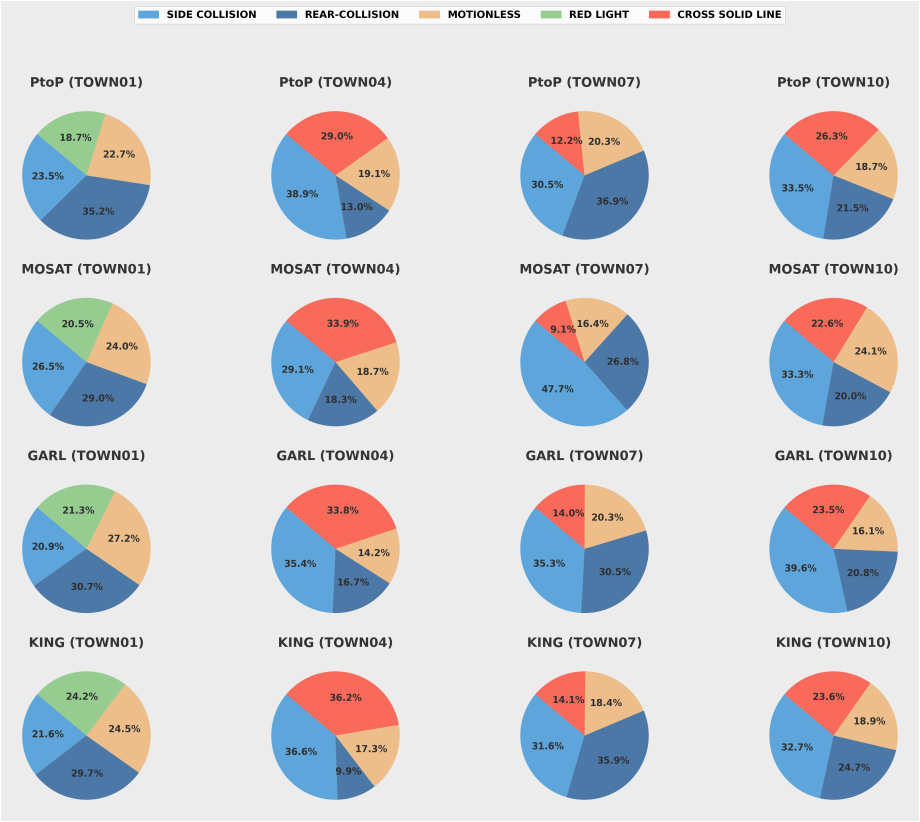

Figure 2: Failure Taxonomy

Table 1: Two-sided paired t-test results for pairwise town comparisons under PtoP (n=4 runs): p-values, 95% confidence intervals (CIs), and effect sizes. CIs and effect sizes are computed for the paired difference (TownA − TownB).

Metric	**Comparison**	***p*-value**	**95% CI**	**Effect**
Violation Rate	TOWN01 vs TOWN04	0.0236	$[0.0121, 0.0829]$	2.133
	TOWN01 vs TOWN07	0.0006	$[-0.0849, -0.0563]$	-7.860
	TOWN01 vs TOWN10	0.1023	$[-0.0119, 0.0769]$	1.164
	TOWN04 vs TOWN07	0.0044	$[-0.1662, -0.0700]$	-3.907
	TOWN04 vs TOWN10	0.2398	$[-0.0476, 0.0176]$	-0.731
	TOWN07 vs TOWN10	0.0075	$[0.0525, 0.1538]$	3.238
TOP-K	TOWN01 vs TOWN04	0.0002	$[-26.4374, -19.5626]$	-10.647
	TOWN01 vs TOWN07	0.7777	$[-28.2726, 23.2726]$	-0.154
	TOWN01 vs TOWN10	0.1277	$[-14.5017, 3.0017]$	-1.045
	TOWN04 vs TOWN07	0.0807	$[-4.6427, 45.6427]$	1.297
	TOWN04 vs TOWN10	0.0125	$[7.0716, 27.4284]$	2.697
	TOWN07 vs TOWN10	0.6390	$[-23.1403, 16.6403]$	-0.260
Parameter Distance	TOWN01 vs TOWN04	0.0010	$[0.0106, 0.0174]$	6.481
	TOWN01 vs TOWN07	0.0029	$[0.0073, 0.0152]$	4.500
	TOWN01 vs TOWN10	0.0249	$[0.0047, 0.0343]$	2.091
	TOWN04 vs TOWN07	0.2045	$[-0.0082, 0.0027]$	-0.808
	TOWN04 vs TOWN10	0.2598	$[-0.0071, 0.0181]$	0.693
	TOWN07 vs TOWN10	0.1979	$[-0.0077, 0.0242]$	0.824
Map Coverage	TOWN01 vs TOWN04	0.0001	$[9.2768, 12.0032]$	12.420
	TOWN01 vs TOWN07	0.0002	$[-55.4674, -41.4926]$	-11.040
	TOWN01 vs TOWN10	0.0002	$[-37.1660, -27.5740]$	-10.740
	TOWN04 vs TOWN07	0.0001	$[-66.2179, -52.0221]$	-13.254
	TOWN04 vs TOWN10	0.0001	$[-47.6891, -38.3309]$	-14.627
	TOWN07 vs TOWN10	0.0040	$[9.7222, 22.4978]$	4.013

5 Additional Statistical Significance Measures

Table 2: Two-sided paired t-test results of **Apollo** vs. **Autoware** on **Safety Violation Rate** (%) per town (n=4): p-values, 95% confidence intervals (CIs), and effect sizes. CIs and effect sizes are computed for the paired difference (Apollo − Autoware).

Metric	**TOWN01**			**TOWN04**			**TOWN07**			**TOWN10**		
	***p*-value**	**95% CI**	**Effect**	***p*-value**	**95% CI**	**Effect**	***p*-value**	**95% CI**	**Effect**	***p*-value**	**95% CI**	**Effect**
Safety Violation Rate (%)	0.01791	$[-36.9387, 21.1287]$	-21.446	0.01246	$[-16.9494, 7.3194]$	-13.565	0.0115	$[-8.8318, -5.5282]$	-39.054	0.00707	$[-21.1335, 3.6435]$	-16.342